

```

17.
18. # Create a list to store results
19. results = []
20.
21. print(f"Testing {n_iterations} random parameter combinations...")
22.
23. # Generate and test random parameter combinations
24. for i in range(n_iterations):
25.     # Generate random parameters
26.     short_ma = np.random.randint(param_space['short_ma'][0], param_space['short_ma'][1])
27.     long_ma = np.random.randint(param_space['long_ma'][0], param_space['long_ma'][1])
28.
29.     # Ensure short_ma is less than long_ma
30.     if short_ma >= long_ma:
31.         continue
32.
33.     # Calculate indicators
34.     data['SMA_Short'] = data['Adj Close'].rolling(window=short_ma).mean()
35.     data['SMA_Long'] = data['Adj Close'].rolling(window=long_ma).mean()
36.
37.     # Generate signals
38.     data['Signal'] = 0
39.     data.loc[data['SMA_Short'] > data['SMA_Long'], 'Signal'] = 1
40.
41.     # Run backtest
42.     backtest_result = backtest_strategy(data)
43.
44.     # Evaluate results
45.     metrics = evaluate_backtest(backtest_result)
46.
47.     # Store results
48.     result = {
49.         'short_ma': short_ma,
50.         'long_ma': long_ma,
51.         **metrics
52.     }
53.     results.append(result)
54.
55. # Convert results to DataFrame

```